Credit Risk Portfolio Analytics — Solution Overview

What the dashboard shows and how a risk analyst uses it

The app covers a **14,685-account** portfolio (**10,000 customers**, $601M exposure, **16.1%** default rate). **Portfolio view:** the analyst sees headline KPIs, model risk-band segments, exposure, and default rate across loan type, employment, region, income and age, plus an employment×income heat-map; loan-type and region filters drive every chart. **Account view:** selecting an account shows its model risk score, risk band, and the top SHAP drivers behind the score for case-level review. The analyst uses the portfolio view to spot concentrations of risk and the account view to understand and action individual exposures.

AI feature 1 — Ask the Portfolio (LLM)

A pre-computed facts pack of exact aggregates is sent to an OpenAI-compatible LLM as grounding; the model answers only from those figures and cites the source table. **Trigger:** analyst submits a question. **False positive:** answering beyond the data — blocked by the strict decline rule. **False negative:** an oddly-phrased valid question isn't matched — the analyst rephrases; the facts pack is always shown. **Human step:** the analyst reads the cited figure and decides whether to act. With no API key the app falls back to a deterministic answerer over the same figures, so numbers are never hallucinated.

AI feature 2 — Anomaly detection

Accounts are scored for unusual repayment/bureau behaviour versus their peer segment (loan type × income band) via peer z-scores plus an IsolationForest; the top 5% are flagged. **Trigger:** composite anomaly score above the 95th percentile. **False positive:** a wealthy borrower who prepays in lumps looks odd but isn't risky. **False negative:** behaviour that mimics normal repayment until a sudden default. **Human step:** each flag carries a plain-English reason and the analyst confirms or dismisses it.

Model results and approach

Logistic regression (transparent baseline) vs XGBoost (challenger), both on engineered bureau + repayment + collections features with no target leakage. **XGBoost: AUC 0.666, Gini 0.332, KS 0.254**; LogReg AUC 0.670, Gini 0.340, KS 0.246. The signal is genuinely subtle (top single-feature correlation ~0.16), so high-0.60s AUC is the honest ceiling; XGBoost leads on KS and stays well-calibrated across bands. Top SHAP drivers: annual income, years employed, repayment days-past-due, age and bureau delinquency.

Top 3 portfolio insights

1. Mortgages are the riskiest product at 24.6% default vs 13.9% for auto loans — so what: tighten mortgage underwriting / pricing despite collateral.

2. Employment type dominates risk: unemployed borrowers default at 35.0% vs 13.6% for business owners — so what: weight employment stability heavily at origination.

3. Cross-cut (employment × income): default risk falls steeply as income rises *within every* employment type — unemployed <25k default at 44.6% vs 20.0% at 100–200k — so what: low income compounds employment risk, so segment limits on the combination, not either alone.

Key assumptions & production needs

Assumptions: target_default is correctly labelled; median/0 imputations are acceptable (flags retained); a fixed reference date anchors recency features. **From the bank's IT/data teams:** a governed feature pipeline with fresh bureau/repayment feeds and agreed SLAs; a secrets-managed LLM endpoint with PII-safe boundaries (only aggregates sent); a model registry with champion/challenger promotion; monitoring hooks for AUC/KS, PSI and calibration drift; and access controls plus an audit log for analyst confirm/dismiss decisions.